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Structured zeroing neural network solution for the mathematical model of a fan dynamic resistor network and its applications

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ABSTRACT

Resistor networks are key tools in interdisciplinary research, however, dynamic resistor networks have not been thoroughly studied. In this study, we introduce a structured zeroing neural network model to tackle the mathematical model of fan dynamic resistor networks, thus filling the research gap in the field of dynamic resistor networks. This study integrates the inherent structural characteristics of special matrices with the core layer in the design of dynamic system algorithms, and accordingly devises a class of fast algorithms for this neural network model. Under identical experimental settings, the computational efficiency of the proposed model is 300–500 times higher than that of conventional methods, and its convergence speed is also improved. The neural network model is further employed to compute the equivalent resistance between any two points in fan resistor networks. Furthermore, this study incorporates the physical property of the natural descent of electric potential into robot path planning, and proposes a robot path planning method tailored to fan-shaped curved surface environments. Compared with traditional path-planning approaches, the proposed method attains higher computational efficiency in such scenarios, as the network scale grows, the advantages of our path planning algorithms become increasingly prominent, highlighting the method's excellent scalability. Finally, we propose two conjectures.

1. Introduction

As a basic physical and mathematical tool, resistor networks possess wide-ranging and significant application value in the fields of circuit design, physics, and computer science. Furthermore, in recent years, their scope of application has been further expanded to more interdisciplinary research fields. Within the field of organic electronics, resistor networks served an indispensable function in exploring defective regions within thin organic films [1]. In the field of materials science, resistor-network-based numerical computation approaches have become common for exploring solid oxide fuel cells [2]. In addition, Andre Geim and Konstantin Novoselov jointly received the Nobel Prize in Physics for their groundbreaking experiments on graphene networks [3,4]. These studies demonstrate that resistor networks play a vital role in the fields of science and engineering.

The research history of resistor networks can be dated back to 1847, Kirchhoff focused the primary study of resistor networks on determining potential distribution and equivalent resistance values [5], laying the foundation for modern research in this field.

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By utilizing sophisticated Laplacian operators, Wu developed a brilliant method in 2004 to determine the resistance between any two nodes in a finite lattice network with a regular structure [6]. However, this approach becomes less effective when extended to irregular lattice structures due to increased complexity. Izmailian and his collaborators successfully overcame the above limitation and proposed a mathematical model of the fan resistor network [7]. In practical engineering and robotic applications, the environmental topology is not always a regular rectangular grid; instead, it more naturally appears as a fan-like structure centered at a reference point and expanding along angular directions. Therefore, studying fan resistor networks is of substantial practical significance. In this paper, we employ a zeroing neural network approach to solve the mathematical model of the fan dynamic resistor network, and further map the resulting potential distribution to an environmental potential field for robot path planning. Neural networks have been vital in promoting artificial intelligence and computer science, emerging as a critical tool for addressing diverse and intricate challenges across scientific and engineering disciplines [8–14]. In 2002, Zhang introduced a novel class of recurrent neural networks known as zeroing neural network (ZNN) [15]. Distinguished by their inherent ability to dynamically track the solution trajectory of time-varying problems [16], ZNNs offer superior computational efficiency and robustness compared to traditional RNNs in handling time-varying problems. Currently, ZNNs have been applied to various fields [17,18], including multi-robot tracking and formation control [19–23], dynamic cryptography [24], medical internet of things (IoT) [25], mathematical computation [26], and financial optimization [27]. These research progresses and applications fully highlight the outstanding performance and broad prospects of ZNNs for solving time-varying problems. The strong capability of ZNNs in solving time-varying problems makes them particularly suitable for handling dynamic resistor networks.

However, despite significant advances in the field of resistor networks, current research remains predominantly concentrated on static resistivity characteristics, while the study of dynamic resistor networks still presents a considerable research gap. This gap severely restricts the potential application of resistor networks in time-varying systems. Therefore, advancing theoretical and methodological research on dynamic resistor networks carries significant academic and practical importance.

The main contributions of this work are summarized as follows:

1. This work introduces a structured zeroing neural network model (SZNNFRN) to solve the mathematical model of fan dynamic resistor networks. Beyond enriching existing solution strategies, it closes a methodological gap in the study of dynamic resistor networks. We demonstrate its use in computing two-point equivalent resistance in a fan network. Compared with traditional methods, the proposed method avoids complex formula derivation while maintaining accuracy, thus providing a new approach for solving such problems.
2. This study integrates the inherent structural characteristics of special matrices with the core layer in the design of dynamic system algorithms, and devises a class of dedicated fast algorithms for SZNNFRN. Results from comparative experiments demonstrate that this fast algorithm can significantly enhance the computational efficiency of the SZNNFRN model. Specifically, the proposed fast algorithm breaks through the efficiency bottleneck of the SZNNFRN model in solving large-scale problems, enabling it to tackle fan dynamic resistor network problems of larger scales and further expanding the application scope of the model.
3. This study incorporates the physical property of the natural descent of electric potential into robot path planning, and develops a path-planning scheme for robots in fan-shaped curved surface environments. Through simulation experiments in static and dynamic environments, the feasibility and effectiveness of the proposed method are fully verified. Comparative analysis with traditional path planning methods shows that this method can reach the target endpoint the fastest in the fan-shaped curved surface environments, highlighting its significant practical significance and broad application prospects.

The rest of this paper is structured as follows. Section 2 introduces the fan resistor network's mathematical model and its equivalent resistance. Section 3 describes the adopted zeroing neural network method. Section 4 develops a fast algorithm for SZNNFRN. Sections 5 and 6 respectively conduct the convergence analysis and numerical experiments of the SZNNFRN model. Section 7 presents two applications: using the SZNNFRN approach to obtain two-node equivalent resistance in fan resistor networks, and robot path planning. Section 8 includes discussions and conjectures. Section 9 provides a conclusion of the entire study.

2. Mathematical model and equivalent resistance of fan resistor network

2.1. Fan resistor network

The fan resistor network with dimensions $m \times n$ is shown in Fig. 1. Where s denotes the longitudinal resistance and r denotes the latitudinal resistance.

According to [7], the fan resistor network's mathematical model can be expressed as

$$L_{mn}^{\text{fan}} V = \mathcal{I}, \tag{1}$$

where

$$V = \left(V_1, V_2, V_3, \dots, V_{mn}\right)^{\top},$$

$$\mathcal{I} = \left(\mathcal{I}_1, \mathcal{I}_2, \mathcal{I}_3, \dots, \mathcal{I}_{mn}\right)^{\top},$$

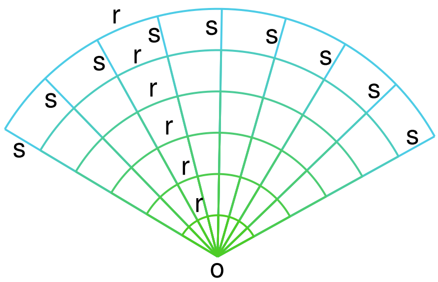


Fig. 1. A fan resistor network with dimensions $m \times n$ (where $m = 6$, $n = 9$).

For the ith node, V_i indicates its potential and $\mathcal{I}_i$ indicates its current. The form of L_{mn}^{fan} is

$$L_{mn}^{\text{fan}} = r^{-1} L_n^{\text{free}} \otimes E_m + s^{-1} E_n \otimes L_m^{\text{DN}}, \tag{2}$$

where E_n and E_m denote the identity matrices of order n and order m respectively, r and s are real numbers.

$$L_n^{\text{free}} = \begin{pmatrix} 1 & -1 & 0 & \cdots & 0 & 0 \\ -1 & 2 & -1 & \ddots & 0 & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & 0 & \ddots & -1 & 2 & -1 \\ 0 & 0 & \cdots & 0 & -1 & 1 \end{pmatrix}_{n\times n}$$

L_n^{free} can be decomposed into $L_n^{\text{per}} + S_n$.

$$L_n^{\text{per}} = \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 & -1 \\ -1 & 2 & -1 & \ddots & \ddots & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & 0 & \ddots & -1 & 2 & -1 \\ -1 & 0 & \cdots & 0 & -1 & 2 \end{pmatrix}_{n\times n},$$

and

$$S_n = \begin{pmatrix} -1 & 0 & \cdots & \cdots & 0 & 1 \\ 0 & & & \ddots & 0 & 0 \\ \vdots & & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & & \vdots \\ 0 & 0 & \ddots & & & 0 \\ 1 & 0 & \cdots & \cdots & 0 & -1 \end{pmatrix}_{n\times n}.$$

$$L_m^{\text{DN}} = \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 & 0 \\ -1 & 2 & -1 & \ddots & \ddots & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & 0 & \ddots & -1 & 2 & -1 \\ 0 & 0 & \cdots & 0 & -1 & 1 \end{pmatrix}_{m\times m}.$$

The current needs to meet

$$\sum_{i=1}^{mn} \mathcal{I}_i = 0.$$

2.2. Fan dynamic resistor network

Most conductors show a temporal variation in resistance [28]. We express the variation of resistance over time as

$$r(t) = r_0\left(1 + \frac{|\sin t|}{q}\right), \ s(t) = s_0\left(1 + \frac{|\sin t|}{q}\right), \quad q \geq 100, \tag{3}$$

where s_0 and r_0 denote the initial resistance.

The fan dynamic resistor network's mathematical model is

$$L_{mn}^{\text{fan}}(t)V(t) = \mathcal{I}, \tag{4}$$

in which

$$L_{mn}^{\text{fan}}(t) = r^{-1}(t)L_n^{\text{free}} \otimes E_m + s^{-1}(t)E_n \otimes L_m^{\text{DN}}, \tag{5}$$

while $V(t)$ stands for the time-varying potential.

2.3. Equivalent resistance

The Laplace matrix (LM) method was first established in [6] for the calculation of large-scale resistor network models. On this basis, [7] improved the LM method to solve the equivalent resistance of the globe network with a zero structure, and [29] improved this method to derive the equivalent resistance of the fan resistor network, where the improved calculation method in [29] is featured with simplicity and convenience in operation. According to [29], excluding point O, the equivalent resistance $R^{\text{fan}}(r_1, r_2)$ between any two points $r_1 = (x_1, y_1)$ and $r_2 = (x_2, y_2)$ in the fan resistor network is given by

$$\begin{aligned} R^{\text{fan}}(r_1, r_2) &= \sum_{m=0}^{M-1}\sum_{n=0}^{N-1} \frac{\left|\psi_{(m,n);(x_1,y_1)}^{\text{fan}} - \psi_{(m,n);(x_2,y_2)}^{\text{fan}}\right|^2}{\lambda_{(m,n)}} \\ &= \frac{2s}{N(2M+1)} \sum_{m=0}^{M-1} \left\{ \sum_{n=1}^{N-1} \frac{2(C_1S_1 - C_2S_2)^2}{h(1-\cos\theta_n) + 1 - \cos 2\varphi_m} + \frac{(S_1 - S_2)^2}{1 - \cos 2\varphi_m} \right\}, \end{aligned} \tag{6}$$

where $h = s/r$ and

$$\begin{aligned} &C_1 = \cos\left((x_1 + 1/2)\theta_n\right), \quad C_2 = \cos\left((x_2 + 1/2)\theta_n\right), \quad \varphi_m = \frac{(m+\frac{1}{2})\pi}{2M+1}, \quad m = 0, 1, \ldots, M-1, \\ &S_1 = \sin\left(2y_1\varphi_m\right), \quad S_2 = \sin\left(2y_2\varphi_m\right), \quad \theta_n = \frac{\pi n}{N}, \quad n = 1, 2, \ldots, N-1. \end{aligned}$$

Specifically, the resistance between apex junction O = (0,0) and the boundary point A = (x,M), can be expressed as [29]

$$\begin{aligned} R^{\text{fan}}(x) &= \frac{s}{N(2M+1)} \sum_{m=0}^{M-1} \left\{ \sum_{n=1}^{N-1} \frac{2\cos^2\left((x+1/2)\theta_n\right)\sin^2\left(2M\varphi_m\right)}{h\sin^2\frac{\theta_n}{2} + \sin^2\varphi_m} + \frac{\sin^2\left(2M\varphi_m\right)}{\sin^2\varphi_m} \right\} \\ &= \frac{sM}{N} + \frac{2s}{N(2M+1)} \sum_{m=0}^{M-1}\sum_{n=1}^{N-1} \frac{\cos^2\left((x+1/2)\theta_n\right)\cos^2\varphi_m}{h\sin^2\frac{\theta_n}{2} + \sin^2\varphi_m}, \end{aligned} \tag{7}$$

where $x \in \{1, \ldots, N\}$.

Essam, Tan and Wu, based on the method in Ref. [30], successfully proposed a single-sum method for solving the equivalent resistance of fan resistor networks [31] i.e.,

$$R_{\text{fan}} = \frac{r}{2m+1} \sum_{i=1}^{m} \frac{\alpha_{\text{fan}} C_i(y_q)^2 - 2\beta_{\text{fan}} C_i(y_q)C_i(y_p) + \gamma_{\text{fan}} C_i(y_p)^2}{\sinh(2L_i)\sinh(2nL_i)}, \tag{8}$$

where $L_i = \frac{1}{2}\log\lambda_i$, $C_i(y) = \cos\big((2y+1)\theta_i\big)$, $\theta_i = \frac{2i-1}{2k+1}\frac{\pi}{2}$, $\upsilon_i = 2\cos(2\theta_i)$, $u_i = 2 + \frac{r}{r_0}(2 - \upsilon_i)$ and

$$\alpha_{\text{fan}} = 4\cosh\big((2t - 2q + 1)L_i\big)\cosh\big((2s + 2q + 1)L_i\big), \quad \beta_{\text{fan}} = 4\cosh\big((2t - 2q + 1)L_i\big)\cosh\big((2s - 2p + 1)L_i\big),$$

$$\gamma_{\text{fan}} = 4\cosh\big((2t + 2p + 1)L_i\big)\cosh\big((2s - 2p + 1)L_i\big).$$

$\lambda_i, \bar{\lambda}_i$ are the greater and lesser solutions of $\lambda_i^2 - u_i\lambda_i + 1 = 0$.

3. The SZNNFRN model

We define the error function of Eq. (4) as

$$E(t) = L_{mn}^{\text{fan}}(t)V(t) - \mathcal{I}, \tag{9}$$

where $V(t) \in \mathbb{R}^{mn}$. When all elements of the error function $E(t)$ equal zero, $V(t)$ represents an exact solution of Eq. (4). In order to guarantee that $\dot{E}(t)$ approaches zero, let

$$\dot{E}(t) = -\Psi(t)\mathcal{F}(E(t)), \tag{10}$$

$\Psi(t)$ serves as a matrix that is positive definite, let $\Psi(t) = \gamma[L_{mn}^{\text{fan}}(t)L_{mn}^{\text{fan}}(t)^{\mathsf{T}} + \lambda E_{mn}]^k$ so as to accelerate the convergence rate of the neural network model. E_{mn} is a unit matrix. $\gamma,\ \lambda,\ k$ need to satisfy $\gamma > 0,\ \lambda \geq 1,\ k \geq 1$.

To sum up, the SZNNFRN for solving the fan dynamic resistor network's mathematical model can be expressed as

$$L_{mn}^{\text{fan}}(t)\dot{V}(t) = -\Psi(t)\mathcal{F}\left(L_{mn}^{\text{fan}}(t)V(t) - \mathcal{I}\right) - \dot{L}_{mn}^{\text{fan}}(t)V(t). \tag{11}$$

$\mathcal{F}(\cdot)$ is the activation function. Below are listed some activation functions suitable for SZNNFRN.

1. Linear activation function (LAF) [32]

$$f(k) = k.$$

2. Bipolar sigmoid activation function (BSAF) [32]

$$f(k) = \frac{1-\exp(-gk)}{1+\exp(-gk)},$$

in which $g \geq 1$.

3. Power sigmoid activation function (PSAF) [32]

$$f(k) = \begin{cases} k^{\upsilon}, & \text{if } |k| \geq 1 \\ \frac{1+\exp(-g)}{1-\exp(-g)} \frac{1-\exp(-gk)}{1+\exp(-gk)}, & \text{if } |k| < 1 \end{cases},$$

in which $g > 2,\ \upsilon = 2h+1,\ h = 1, 2\ldots$

4. Smooth power sigmoid function (SPSF) [32]

$$f(k) = \frac{1}{2}k^{\upsilon} + \frac{1+\exp(-g)}{1-\exp(-g)} \frac{1-\exp(-gk)}{1+\exp(-gk)},$$

in which $g > 2,\ \upsilon = 2h+1,\ h = 1, 2\ldots$

5. Hyperbolic sine activation function (HSAF) [33]

$$f(k) = \frac{1}{2}(\exp(gk) - \exp(-gk)),$$

in which $g > 1$.

6. Hyperbolic Tangent (Tanh) [34]

$$f(k) = \tanh(k).$$

7. Mish [34]

$$f(k) = k \cdot \tanh(\ln(1+e^{k})).$$

Remark 1. According to Eq. (11), the ith neuron's dynamics (for $i = 1, 2, \ldots, mn$) in the SZNNFRN model may be expressed in the form of

$$\dot{V}_i = \sum_{j=1}^{mn} p_{i,j}\dot{V}_i - \gamma \sum_{j=1}^{mn} \alpha_{i,j} f_j - \sum_{j=1}^{mn} u_{i,j} V_j, \tag{12}$$

where

$$f_j = f\left(\sum_{k=1}^{mn} \delta_{j,k} v_k - b_j\right), \tag{13}$$

V_i represents the ith component of the $V(t)$, which corresponds to neuron i in Eq. (11). $p_{i,j}$, $\alpha_{i,j}$, and $u_{i,j}$ denote the i,jth entries of matrices $E_{mn} - L_{mn}^{\text{fan}}(t)$, $\left[L_{mn}^{\text{fan}}(t)L_{mn}^{\text{fan}}(t)^{\mathsf{T}} + \lambda E_{mn}\right]^{k}$, and $\dot{L}_{mn}^{\text{fan}}(t)$, in corresponding order. The symbol $\delta_{j,k}$ denotes the j,kth entry of $L_{mn}^{\text{fan}}(t)$. The term b_j stands for the jth entry of $\mathcal{I}_{mn}$. The function $f(\cdot)$ acts element-wise as the scalar form of $\mathcal{F}(\cdot)$. A schematic of the SZNNFRN network corresponding to (11) is provided in Fig. 2.

4. Fast algorithm for SZNNFRN

With the continuous expansion of problem scale and the growing demand for large-scale data processing [35,36], the development of more efficient and targeted algorithms has become increasingly crucial. Building on fast algorithms tailored to matrices with a specific structure [37–47], we leverages the unique structural properties of $L_{mn}^{\text{fan}}(t)$ to propose a class of fast algorithm tailored for the SZNNFRN model. Experimental results show that under exactly the same experimental conditions, the proposed algorithm in this paper significantly improves the computational efficiency of the model while maintaining computational accuracy, and achieves a 300–500 times speedup across all problem scales investigated in this study, thus effectively saving computational resources. Moreover, the proposed fast algorithm not only improves the scalability of the SZNNFRN model but also provides strong support for its potential applications in future data-intensive scenarios.

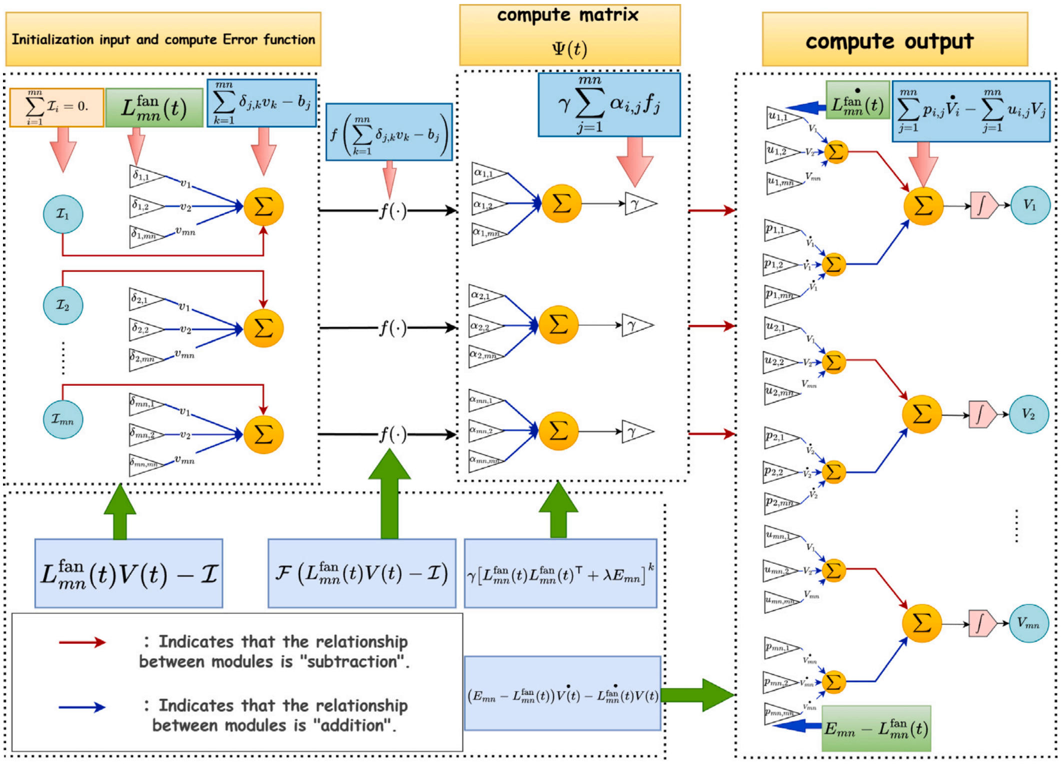


Fig. 2. SZNNFRN model network structure diagram.

4.1. Fast algorithm for $L^{\text{fan}}_{mn}(t)V(t)$

At the instant t^*, $v = V(t^*) = [V_1(t), V_2(t), \dots, V_{mn}(t)]^{\mathsf{T}}$, $L^{\text{fan}}_{mn}v$ can be decomposed into two items for calculation, $(L^{\text{free}}_n \otimes E_m)v$ and $(E_n \otimes L^{\text{DN}}_m)v$, and $(L^{\text{free}}_n \otimes E_m)$ can be further split into $(L^{\text{per}}_n \otimes E_m + S_n \otimes E_m)$. $S_n \otimes E_m$ is a sparse matrix where most elements are zero, the computational speed of $(S_n \otimes E_m)v$ is relatively fast. $L^{\text{per}}_n \otimes E_m$ forms a circulant matrix, and is determined through its first row (denoted as h), h has the form of

$$h = \begin{cases} [2, \underbrace{0, \dots, 0}_{m-1}, -1, \underbrace{0, \dots, 0}_{m-1}]^T, & n = 2, \\ [2, \underbrace{0, \dots, 0}_{m-1}, -1, \underbrace{0, \dots, 0}_{mn-2\ m-1}, -1, \underbrace{0, \dots, 0}_{m-1}]^T, & n > 2. \end{cases}$$

The fast algorithm targeting $(L^{\text{per}}_n \otimes E_m)v$ is elaborated in Algorithm 1.

Algorithm 1 Fast algorithm for $(L^{\text{per}}_n \otimes E_m)v$

Step 1: Compute $\omega = \text{FFT}(v)$;
Step 2: Compute $d = \text{FFT}(h)$;
Step 3: Compute the element wise vector–vector product $\varpi = \omega \cdot d$;
Step 4: Compute $(L^{\text{per}}_n \otimes E_m)v = \text{IFFT}(\varpi)$.

$(E_n \otimes L^{\text{DN}}_m)v$ takes the specific form of

$$(E_n \otimes L^{\text{DN}}_m)v = \begin{bmatrix} L^{DN}_m \zeta_1 & 0 & \cdots & 0 \\ 0 & L^{DN}_m \zeta_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & L^{DN}_m \zeta_n \end{bmatrix},$$

in which $\zeta_i = [v_{(i-1)m+1}, \dots, v_{(i-1)m+m}]$, $i = 1, 2, 3 \dots, n$. Once $z_i = L^{\text{DN}}_m \zeta_i$ is solved, $(E_n \otimes L^{\text{DN}}_m)v$ is able to be derived. The fast algorithm targeting $(E_n \otimes L^{\text{DN}}_m)v$ is elaborated in Algorithm 2.

Algorithm 2 Fast algorithm for $(E_n \otimes L_m^{\mathrm{DN}})v$

Step 1: Compute $z_i(1) = 2\zeta_i(1) - \zeta_i(2)$;
Step 2: Cycle computing $z_i(j) = -\zeta_i(j-1) + 2\zeta_i(j) - \zeta_i(j+1),\ j = 2, 3, \dots, m-1$;
Step 3: Compute $z_i(m) = -\zeta_i(m-1) + \zeta_i(m)$;
Step 4: Concatenate n independent sub-vectors z_i, $(I_n \otimes L_m^{\mathrm{DN}})v = [z_1^T, z_2^T, \dots, z_n^T]^T$.

Based on the two fast algorithms and the expansion of Eq. (5)

$$\begin{aligned}L_{mn}^{\mathrm{fan}}(t)V(t) &= r^{-1}(t)(L_n^{\mathrm{free}} \otimes E_m)V(t) + s^{-1}(t)(E_n \otimes L_m^{\mathrm{DN}})V(t)\\ &= r^{-1}(t)\left[(L_n^{\mathrm{per}} + S_n) \otimes E_m\right]V(t) + s^{-1}(t)(E_n \otimes L_m^{\mathrm{DN}})V(t)\\ &= r^{-1}(t)\left[(L_n^{\mathrm{per}} \otimes E_m)V(t) + (S_n \otimes E_m)V(t)\right] + s^{-1}(t)(E_n \otimes L_m^{\mathrm{DN}})V(t).\end{aligned}$$

Then, $L_{mn}^{\mathrm{fan}}(t)V(t)$ can be evaluated with high efficiency. Similarly, computing $\dot{L}_{mn}^{\mathrm{fan}}(t)V(t)$ calls for an initial differentiation of $L_{mn}^{\mathrm{fan}}(t)$, the rest proceeds as before.

4.2. Fast algorithm for $\left[L_{mn}^{\mathrm{fan}}(t)L_{mn}^{\mathrm{fan}}(t)^{\mathsf{T}} + \lambda E_{mn}\right]^k \mathcal{F}(\cdot)$

$L_{mn}^{\mathrm{fan}}(t)$ can be simplified into a diagonal matrix through the action of the type VI discrete sine transform and the type III discrete cosine transform.

$$\begin{aligned}&(\mathbb{C}_n^{III} \otimes \mathbb{S}_m^{VI})^{-1} L_{mn}^{\mathrm{fan}}(\mathbb{C}_n^{III} \otimes \mathbb{S}_m^{VI})\\ &= ((\mathbb{C}_n^{III})^{-1} \otimes (\mathbb{S}_m^{VI})^{-1})(r^{-1}L_n^{\mathrm{free}} \otimes E_m + s^{-1}E_n \otimes L_m^{\mathrm{DN}})(\mathbb{C}_n^{III} \otimes \mathbb{S}_m^{VI})\\ &= r^{-1}((\mathbb{C}_n^{III})^{-1}L_n^{\mathrm{free}}\mathbb{C}_n^{III}) \otimes ((\mathbb{S}_m^{VI})^{-1}\mathbb{S}_m^{VI}) + s^{-1}((\mathbb{C}_n^{III})^{-1}\mathbb{C}_n^{III}) \otimes ((\mathbb{S}_m^{VI})^{-1}L_m^{\mathrm{DN}}\mathbb{S}_m^{VI})\\ &= r^{-1}\Lambda_n \otimes E_m + s^{-1}E_n \otimes \Lambda_m\\ &= \Lambda_{mn},\end{aligned} \tag{14}$$

where S_m^{VI} is the type VI discrete sine transform (DST-VI) [48],

$$S_m^{VI} = \frac{1}{\sqrt{2\ m+1}}\left(\sin\frac{(2j-1)\,k\pi}{2\ m+1}\right)_{j,k=1}^{m},$$

C_n^{III} is the type III discrete cosine transform (DCT-III) [48–50],

$$\mathbb{C}_n^{III} = \left(\sqrt{\frac{2}{n}}d_k\cos\frac{(k-1)(2j-1)\pi}{2n}\right)_{k,j=1}^{n},$$

$$d_k = \min\left\{1, \frac{\sqrt{2}}{2} - 1 + k\right\}.$$

L_n^{free} can be brought by similarity to form Λ_n, and L_m^{DN} to Λ_m.

$$\Lambda_n = \mathrm{diag}(\lambda_0, \lambda_1, \dots, \lambda_j, \dots, \lambda_{n-1}), \quad \lambda_j = 2 - 2\cos\left(\frac{j\pi}{n}\right), \quad j = 0, 1, \dots, n-1,$$

$$\Lambda_m = \mathrm{diag}(\lambda'_0, \lambda'_1, \dots, \lambda'_k, \dots, \lambda'_{m-1}), \quad \lambda'_k = 2 - 2\cos\left(\frac{(2k+1)\pi}{2m+1}\right), \quad k = 0, 1, \dots, m-1,$$

$L_{mn}^{\mathrm{fan}}(t)$ can be brought to the diagonal matrix Λ_{mn},

$$\Lambda_{mn}(t) = \mathrm{diag}(\lambda_{0,0}(t), \lambda_{0,1}(t), \dots, \lambda_{n-1,m-1}(t)), \quad \lambda_{j,k} = r^{-1}(t)\lambda_j + s^{-1}(t)\lambda'_k.$$

By Eq. (14), we can obtain

$$\begin{aligned}L_{mn}^{\mathrm{fan}}(t) &= (\mathbb{C}_n^{III} \otimes \mathbb{S}_m^{VI})\Lambda_{mn}(t)(\mathbb{C}_n^{III} \otimes \mathbb{S}_m^{VI})^{-1}\\ &= (\mathbb{C}_n^{III} \otimes E_m)(E_n \otimes \mathbb{S}_m^{VI})\Lambda_{mn}(t)((\mathbb{C}_n^{III})^{-1} \otimes E_m)(E_n \otimes (\mathbb{S}_m^{VI})^{-1}).\end{aligned} \tag{15}$$

Since L_{mn}^{fan} is diagonalizable, $L = L^{\mathsf{T}}$ and therefore $L_{mn}^{\mathrm{fan}}L_{mn}^{\mathrm{fan}\,\mathsf{T}} = (L_{mn}^{\mathrm{fan}})^2$. Then, we have

$$\left[(L_{mn}^{\mathrm{fan}})^2 + \lambda E_{mn}\right]^k = (\mathbb{C}_n^{III} \otimes E_m)(E_n \otimes \mathbb{S}_m^{VI})M(t)^k((\mathbb{C}_n^{III})^{-1} \otimes E_m)(E_n \otimes (\mathbb{S}_m^{VI})^{-1}),$$

with

$$M(t) = \begin{bmatrix}\lambda_{0,0}^2(t) + \lambda & 0 & \cdots & 0\\ 0 & \lambda_{0,1}^2(t) + \lambda & \ddots & \vdots\\ \vdots & \ddots & \ddots & 0\\ 0 & \cdots & 0 & \lambda_{n-1,m-1}^2(t) + \lambda\end{bmatrix}.$$

To compute $\left[L_{mn}^{\text{fan}} L_{mn}^{\text{fan}\,\mathsf{T}} + \lambda E_{mn}\right]^k \mathcal{F}(\cdot)$, first compute $\delta = \left(E_n \otimes \left(S_m^{VI}\right)^{-1}\right)\mathcal{F}(\cdot)$, where

$$\delta = \begin{bmatrix} \left(S_m^{VI}\right)^{-1}\chi_1 & 0 & \cdots & 0 \\ 0 & \left(S_m^{VI}\right)^{-1}\chi_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \left(S_m^{VI}\right)^{-1}\chi_n \end{bmatrix},$$

with

$$\chi_j = [f_{j+1}, \dots, f_{j+m}], \quad j = 1, 2, 3 \cdots n. \tag{16}$$

After computing $\delta_{mn} = E_n \otimes \left(S_m^{VI}\right)^{-1}\chi_j$, the vector $\left(E_n \otimes \left(S_m^{VI}\right)^{-1}\right)\mathcal{F}(\cdot)$ is obtained by orderly merging the n components. $\left(S_m^{VI}\right)^{-1}\chi_j$ is efficiently handled by IDST-VI (type VI inverse discrete sine transform).

$$\begin{aligned} &((\mathbb{C}_n^{III})^{-1} \otimes E_m)\delta_{mn} = \left((\mathbb{C}_n^{III})^{-1} \otimes E_m\right)\left(e_{n,1} \otimes \delta_{1,n} + e_{n,2} \otimes \delta_{2,n} + \cdots + e_{n,m} \otimes \delta_{m,n}\right) \\ &= \left((\mathbb{C}_n^{III})^{-1} e_{n,1} \otimes \delta_{1,n}\right) + \left((\mathbb{C}_n^{III})^{-1} e_{n,2} \otimes \delta_{2,n}\right) + \cdots + \left((\mathbb{C}_n^{III})^{-1} e_{n,m} \otimes \delta_{m,n}\right). \end{aligned} \tag{17}$$

Among them, $e_{n,j}$ is a set of n-dimensional linearly independent unit vector. Notably, the product $(\mathbb{C}_n^{III})^{-1} e_{n,j}$ gives the jth column of $(\mathbb{C}_n^{III})^{-1}$, and this computation will be highly efficient. Let b represents the vector after processing. Consequently, $M(t)^k b$ yields a vector obtained via element-wise multiplication of two vectors.

Likewise, by adopting the vector decomposition approach given in Eq. (16), we perform the sixth kind of discrete sine transform and arrange the resulting vectors sequentially to obtain a new vector ψ. Similarly, following the approach in Eq. (17), we obtain

$$\begin{aligned} (\mathbb{C}_n^{III} \otimes E_m)\psi_{mn} &= \left(\mathbb{C}_n^{III} \otimes E_m\right)\left(e_{n,1} \otimes \psi_{1,n} + e_{n,2} \otimes \psi_{2,n} + \cdots + e_{n,m} \otimes \psi_{m,n}\right) \\ &= \left(\mathbb{C}_n^{III} e_{n,1} \otimes \psi_{1,n}\right) + \left(\mathbb{C}_n^{III} e_{n,2} \otimes \psi_{2,n}\right) + \cdots + \left(\mathbb{C}_n^{III} e_{n,m} \otimes \psi_{m,n}\right). \end{aligned} \tag{18}$$

Similarly, the product $(\mathbb{C}_n^{III} e_{n,j})$ gives the jth column of $\mathbb{C}_n^{III}$. Therefore, the rapid method for calculating $\left[L_{mn}^{\text{fan}}(t) L_{mn}^{\text{fan}}(t)^{\mathsf{T}} + \lambda E_{mn}\right]^k \mathcal{F}(\cdot)$ has been designed and completed.

Algorithm 3 Fast algorithm for $\left[L_{mn}^{\text{fan}}(t) L_{mn}^{\text{fan}}(t)^{\mathsf{T}} + \lambda E_{mn}\right]^k \mathcal{F}(\cdot)$

Step 1: Compute $\delta_j = \text{IDST-VI}(\chi_j)$, where $\chi_j = [f_{j+1}, f_{j+2}, \cdots, f_{j+m}]$, $\quad j = 1, 2, 3 \cdots n$;
Step 2: Concatenate n independent sub-vectors δ_j, $\quad \delta = [\delta_1{}^{\mathsf{T}}, \delta_2{}^{\mathsf{T}}, \dots, \delta_n{}^{\mathsf{T}}]^{\mathsf{T}}$;
Step 3: Compute $b_j = \text{IDCT-III}(\vartheta_j)$, in which $\vartheta_j = [\delta_j, \delta_{j+m}, \cdots, \delta_{j+(n-1)m}]$, $\quad j = 1, 2, 3 \cdots m$;
Step 4: Splice the n sub-vectors b_j in order to get the b;
Step 5: Compute the element wise vector–vector product $\kappa = \mathrm{M^k(t)} \cdot b$;
Step 6: Compute $\psi_j = \text{DST-VI}(\theta_j)$, where $\theta_j = [\kappa_{j+1}, \kappa_{j+2}, \cdots, \kappa_{j+m}]$, $\quad j = 1, 2, \cdots n$;
Step 7: Concatenate n independent sub-vectors ς_j, $\quad \varsigma = [\varsigma_1{}^{\mathsf{T}}, \varsigma_2{}^{\mathsf{T}}, \dots, \varsigma_n{}^{\mathsf{T}}]^{\mathsf{T}}$;
Step 8: Compute $\Upsilon_j = \text{DCT-III}(\iota_j)$, where $\iota_j = [\psi_j, \psi_{j+m}, \cdots, \psi_{j+(n-1)m}]$, $\quad j = \overline{1, m}$;
Step 9: Splice the n sub-vectors Υ_j in order to get the $\left[L_{mn}^{\text{fan}}(t) L_{mn}^{\text{fan}}(t)^{\mathsf{T}} + \lambda E_{mn}\right]^k \mathcal{F}(\cdot)$.

4.3. Comparison of computational efficiency

In Algorithm 1, FFT and IFFT operations constitute the dominant term in the computational complexity, and both FFT and IFFT of length N ($N = mn$) can be computed in $O(N \log_2 N)$ time. The remaining steps (element-wise vector multiplication) only contribute a linear complexity of $O(N)$. Ultimately, Algorithm 1 establishes a quantitative relationship of $O(N \log_2 N)$ between the total number of operations and the dimension N. Compared with the $O(N^2)$ complexity required for direct computation, a significant reduction is achieved.

Algorithm 2 exploits the structural properties of $(E_n \otimes L_m^{\mathrm{DN}})$ to optimize the computational complexity to a linear $O(N)$. This linear $O(N)$ complexity is a direct consequence of the unique matrix structure of $(E_n \otimes L_m^{\mathrm{DN}})$: unlike dense matrices that require $O(N^2)$ operations, $(E_n \otimes L_m^{\mathrm{DN}})$ has non-zero elements only on the main diagonal, the first superdiagonal, and the first subdiagonal. By fully leveraging this extreme sparsity, Algorithm 2 ultimately establishes a strict linear quantitative relationship between the total number of operations and the dimension N.

The overall computational complexity of Algorithm 3 is $O(N \log_2 N)$, which is dominated by the Type III Discrete Cosine Transform (DCT-III), Type VI Discrete Sine Transform (DST-VI), and their inverse transforms. For sequences of length N, transforms such as DCT-III and DST-VI can be computed in $O(N \log_2 N)$ time [50,51]. Since Algorithm 3 requires one round of forward DCT-III and DST-VI transforms as well as one round of inverse DCT-III and DST-VI transforms, these operations constitute the dominant term in the time complexity, leading to an overall complexity of $O(N \log_2 N)$, the remaining steps (element-wise vector multiplication and sub-vector concatenation) only contribute a linear complexity of $O(N)$. Compared with the original complexity of $O(N^3)$ when directly computing the target expression $\left[L_{mn}^{\text{fan}}(t) L_{mn}^{\text{fan}}(t)^{\mathsf{T}} + \lambda E_{mn}\right]^k \mathcal{F}(\cdot)$, Algorithm 3 reduces the computational cost to $O(N \log_2 N)$

Table 1
Specific parameters for simulation settings.

Model	The parameters of the model	Simulation time
ZNN	$\gamma = 3$	[0, 3]
SAND	$\gamma = 3,\ \lambda = 2$	[0, 5]
NNN	$\gamma = 3$	[0, 3]
IMRNN	$\gamma = 3,\ \lambda = 2,\ k = 2$	[0, 0.3]
SZNNFRN	$\gamma = 3,\ \lambda = 2,\ k = 2$	[0, 0.3]

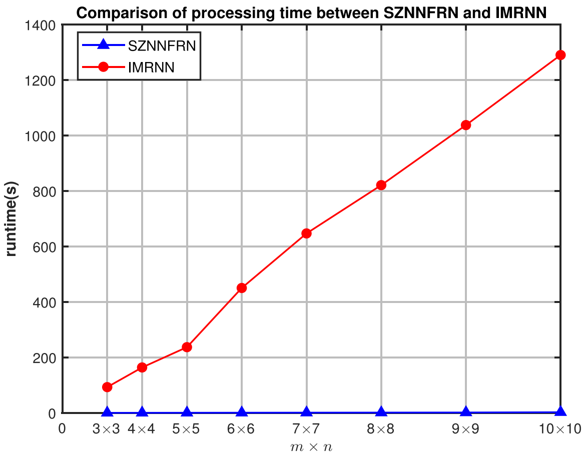


Fig. 3. Comparing the CPU runtime of the SZNNFRN and IMRNN.

by leveraging the structural properties of matrices and the discrete sine and cosine transforms, thereby enabling fast and efficient computation.

We performed systematic numerical simulations using five models: SZNNFRN, IMRNN [32], ZNN [15], NNN [52], and SAND [53] on resistor networks of varying scales. The specific parameter configurations for the numerical experiments are detailed in Table 1. Fig. 3 presents a comparison of CPU runtime between the SZNNFRN and IMRNN models, while Fig. 4 further illustrates the performance comparison of the SZNNFRN model with the other three models (ZNN, NNN, and SAND). Fig. 5 shows a comparison of the precise running times of the five models on fan resistor networks of different scales. When the model's runtime or memory usage exceeds a predefined threshold, the data is marked as "NaN", and the corresponding data points are not displayed in the figure. It is important to note that the simulation duration for each model was selected at the point when it achieved sufficient convergence and reached a stable state, ensuring the reliability of the results. In the simulations, we set $r(t) = s(t) = 1 + 0.004\ t$ to simulate the dynamic characteristics of the time-varying resistor network.

Analysis results of Fig. 3 show that under the conditions of exactly identical hardware configuration (Intel Core i7-13650HX processor (2.60 GHz)), solution accuracy, parameter settings and operating environment, the SZNNFRN algorithm significantly outperforms the IMRNN algorithm in computational efficiency and achieves a 300–500 times acceleration across all problem scales. Notably, as the network scale increases, SZNNFRN demonstrates an exponentially growing computational speed advantage, highlighting its excellent scalability in handling large-scale resistor network problems.

Analysis of Fig. 4 indicates that, across resistor networks of varying scales, SZNNFRN consistently outperforms the other models in computational speed and efficiency. Notably, SZNNFRN's efficiency advantage becomes more pronounced as the problem scale increases. As the network size increases from 10×10 to 15×15, the computation time of the other models rises sharply, while SZNNFRN exhibits a comparatively stable and gradual growth, highlighting its scalability advantage. All experiments were conducted on a unified hardware platform (Intel Core i7-13650 HX processor (2.60 GHz)), using identical numerical precision standards and implemented with the same programming languages, ensuring the fairness of the simulations.

Based on a comprehensive analysis of Figs. 3, 4 and 5, the conclusion can be derived: The SZNNFRN model consistently outperformed the other models in computational efficiency. While the runtimes of ZNN, SAND, NNN, and IMRNN increased sharply with network size, SZNNFRN exhibited a gradual and stable rise in computation time, highlighting its superior efficiency and scalability for large-scale resistor network problems.

5. Convergence analysis

Theorem 1. *Given the resistor network lattice $L_{mn}^{\mathrm{fan}}(t) \in \mathbb{R}^{mn}$ and the current vector $\mathcal{I} \in \mathbb{R}^{mn}$, if the activation function is odd and monotonically increasing (which serves as a sufficient condition for the analysis), then starting from any initial condition $V(t_0)$, the state variable $V(t)$ of SZNNFRN globally converges to $V^*(t) = L_{mn}^{\mathrm{fan}^{-1}}(t)\mathcal{I}$.*

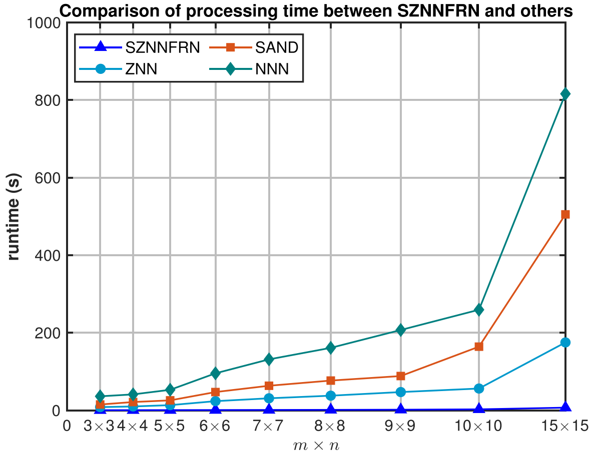


Fig. 4. Comparing the CPU runtime of the SZNNFRN and others.

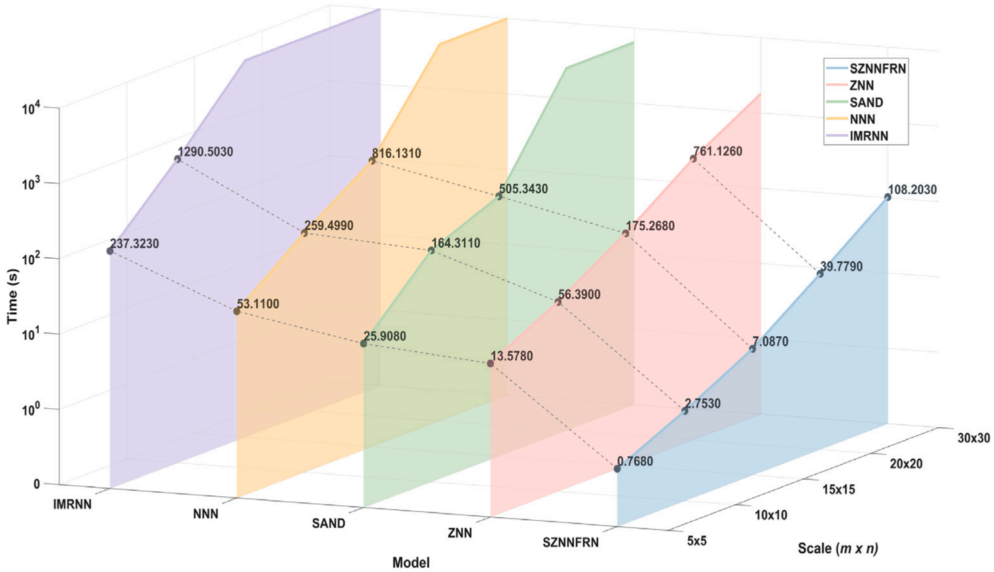


Fig. 5. CPU runtimes of each model for resistor networks at different scales.

Proof. Under the given fan dynamic lattice and current vector, define the error function as $V(t) - L_{mn}^{\mathrm{fan}^{-1}}(t)\mathcal{I}$. When $E(t)$ converges to zero, $V(t)$ approaches $V^*(t)$.

Through Eq. (15) and

$$\frac{\mathrm{d}E(t)}{\mathrm{d}t} = L_{mn}^{\mathrm{fan}}(t)\frac{\mathrm{d}V(t)}{\mathrm{d}t},$$

can derived

$$\dot{E}(t) = -\gamma(L_{mn}^{\mathrm{fan}}(t)L_{mn}^{\mathrm{fan}^{\mathrm{T}}}(t) + \lambda E_{mn})^k \mathcal{F}(E(t)).$$

Let us set the Lyapunov candidate function to be

$$L(E(t),t) = \frac{1}{2}\,\|E(t)\|_F^2 = \frac{1}{2}\mathrm{tr}\left(E(t)^{\mathrm{T}}E(t)\right).$$

If $E(t) \neq 0$, we can obtain that

$$\begin{aligned}\frac{\mathrm{d}L(E(t),t)}{\mathrm{d}t} &= \frac{1}{2}\frac{\mathrm{d}\,\mathrm{tr}\left((E(t))^{\mathrm{T}}(E(t))\right)}{\mathrm{d}t}\\ &= \mathrm{tr}\left((E(t))^{\mathrm{T}}\left(\dot{E}(t)\right)\right)\end{aligned}$$

$$= -\gamma \operatorname{tr}\left(\left(L_{mn}^{\text{fan}}(t) L_{mn}^{\text{fan}\top}(t) + \lambda E_{mn}\right)^{k} (E(t))^{\top} \mathcal{F}(E(t))\right). \tag{19}$$

In the case that A serves as a symmetric matrix that is positive semi-definite, and B is a symmetric matrix, then

$$\lambda_{\min}(A)\operatorname{tr}(B) \leq \operatorname{tr}(AB) \leq \lambda_{\max}(A)\operatorname{tr}(B),$$

where $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ are the minimum and maximum eigenvalues of matrix A, respectively. Since $L_{mn}^{\text{fan}}(t)$ and its transpose have identical spectra, it follows that

$$\begin{aligned}\frac{\mathrm{d}L(E(t),t)}{dt} &\leq -\gamma\left(\alpha^2(t)+\lambda\right)^k \operatorname{tr}\left((E(t))^{\top}\mathcal{F}(E(t))\right)\\ &= -\gamma\left(\alpha^2(t)+\lambda\right)^k \sum_{j=1}^{m}\sum_{i=1}^{n} e_{ij} f(e_{ij}),\end{aligned} \tag{20}$$

where $\alpha(t) = \lambda_{\min}(L_{mn}^{\text{fan}}(t))$.

In the case of the activation function that is monotonically increasing and odd, let its corresponding scalar (elementwise) mapping be denoted by $f(\cdot)$. Accordingly, $f(\cdot)$ needs to satisfy

$$f(k)\begin{cases} >0, & k>0,\\ =0, & k=0,\\ <0, & k<0.\end{cases}$$

Let $e(\cdot)$ denote the elementwise counterpart of $E(\cdot)$. Thus we may deduce that

$$e_{ij} f(e_{ij})\begin{cases} >0, & \text{if } e_{ij}\neq 0,\\ =0, & \text{if } e_{ij}=0.\end{cases}$$

In conclusion

$$L(E(t),t)\begin{cases} <0, & E(t)\neq 0,\\ =0, & E(t)=0.\end{cases}$$

Based on the above analysis, it can be concluded that the error function converges to zero regardless of the initial conditions.

In the case that a LAF is employed, Eq. (19) can be expressed as

$$\begin{aligned}\frac{\mathrm{d}L(E(t),t)}{\mathrm{d}t} &= -\gamma \operatorname{tr}\left(\left(L_{mn}^{\text{fan}}(t) L_{mn}^{\text{fan}\top}(t) + \lambda E_{mn}\right)^{k} (E(t))^{\top}(E(t))\right)\\ &\leq -\gamma(\alpha^2(t)+\lambda)^k \operatorname{tr}\left((E(t))^{\top}(E(t))\right)\\ &= -2\gamma(\alpha^2(t)+\lambda)^k L(E(t),t).\end{aligned}$$

From the analysis of this differential inequality, we obtain that

$$L(t) \leq L(0)\exp(-2\gamma(\alpha^2(t)+\lambda)^k t).$$

With a linear activation, the SZNNFRN model attains a convergence speed of

$$ECR = \gamma\left(\alpha^2(t)+\lambda\right)^k. \quad \square$$

Theorem 2. *Under the setting $\mathcal{F}(x) = x\cdot \Upsilon(x)$ with $\Upsilon(x) > 0$, for the lattice $L_{mn}^{\text{fan}}(t)$ and current vector $\mathcal{I}$, $V(t)$ globally converges, regardless of the random initialization $V(0)$, to the solution of Eq. (15): $V^*(t) = L_{mn}^{\text{fan}^{-1}}(t)\mathcal{I}$.*

Proof. Define the elementwise function of the activation $\mathcal{F}(\cdot)$ as $f(\cdot) = x\cdot v(x)$, with $v(x) > 0$ as well as being the elementwise form of $\Upsilon(x)$. Based on Eq. (20), one obtains

$$\frac{dL(E(t),t)}{dt} \leq -\gamma\left(\alpha^2(t)+\lambda\right)^k \sum_{j=1}^{m}\sum_{i=1}^{n} e_{ij}^2 v\left(e_{ij}\right). \tag{21}$$

Thus, we infer that

$$L(E(t),t)\begin{cases} <0 & E(t)\neq 0,\\ =0 & E(t)=0.\end{cases}$$

In summary, it can be concluded that for all initial states, the error function converges to zero. $\square$

6. Numerical simulations

In this section, numerical simulations of the SZNNFRN model are conducted, which helps to verify the rationality of the model and optimize its parameters.

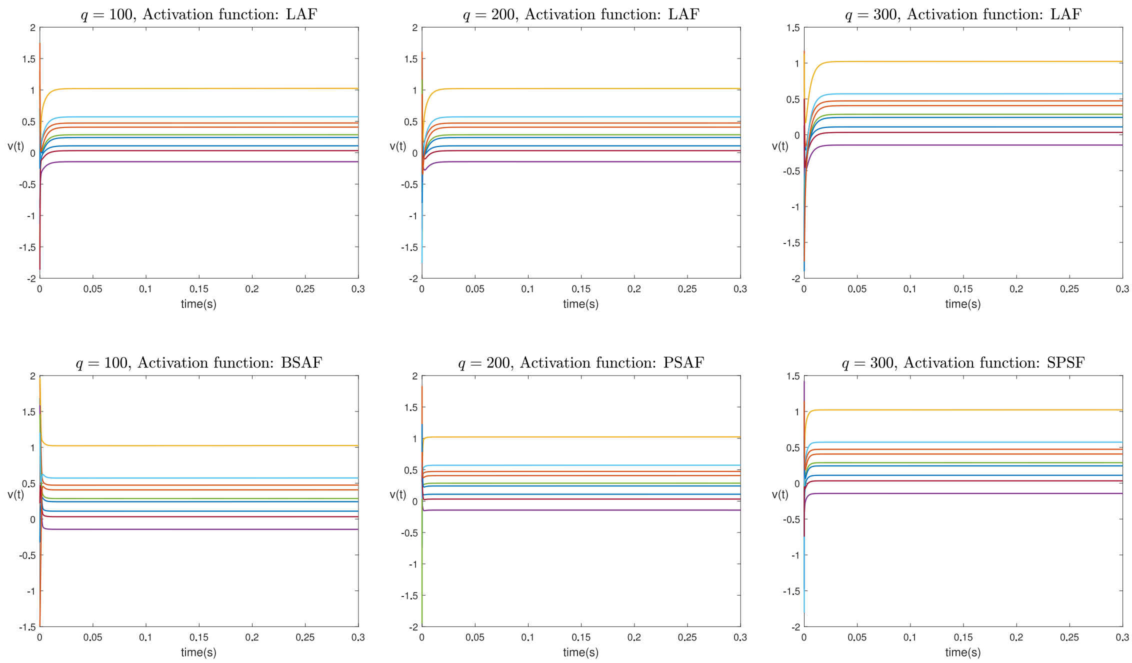


Fig. 6. The trajectory curve of the function $V(t)$ over time.

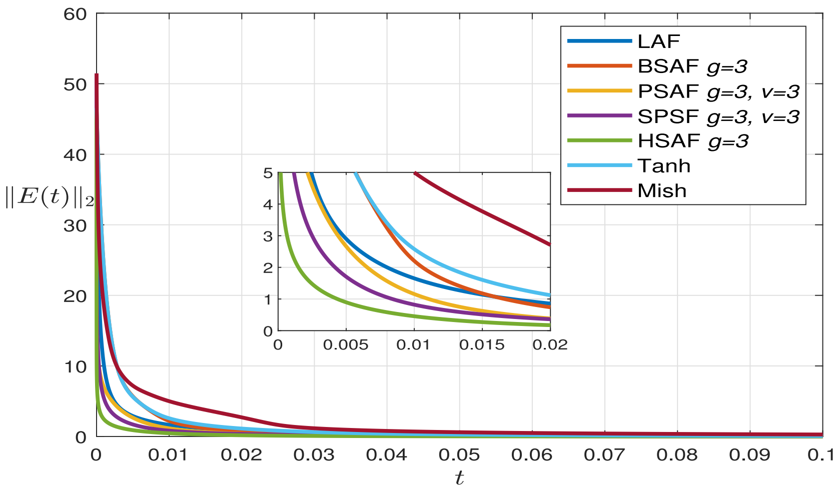
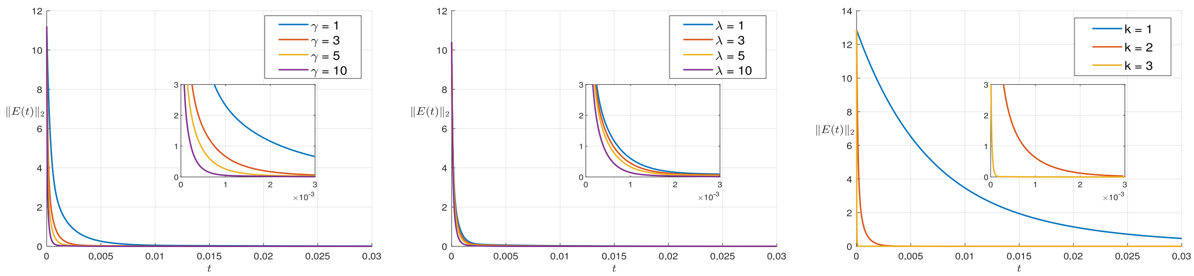


Fig. 7. Path curves of $\|L_{mn}^{\mathrm{fan}}(t)V(t) - \mathcal{I}\|_2$ against time t under different activation functions.

Fig. 6 presents the time-domain variation of $V(t)$ under different activation functions and q values in a 3×3 fan resistor network. The nine curves in Fig. 6 depict the variation of $V(t)$ with time. Analysis of the figure shows that Under different activation functions and q values, the potential curves all exhibit consistency, which reflects the stability of the proposed model to a certain extent.

Fig. 7 presents the comparison of the convergence performance of the error function $E(t)$ under different activation functions in a 10×10 fan resistor network, where $r(t) = s(t) = 1 + \frac{|\sin t|}{100}$. The settings of the model parameters are identical. Fig. 8 illustrates the influence of model parameters γ, λ, and k on convergence performance in a 3×3 fan resistor network. All other conditions are identical except for the differences in model parameters. Specific parameter settings are reflected in the respective subfigures.



(a) (b) (c)

Fig. 8. The influence of model parameters on convergence performance.

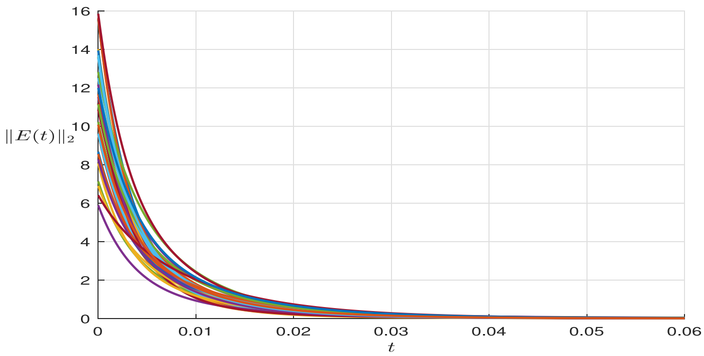


Fig. 9. Curves of $\|L_{mn}^{\text{fan}}(t)V(t)-\mathcal{I}\|_2$ versus time t under various initial states.

Analysis of Figs. 7 and 8 reveals that when HSAF, PSAF and SPSF are used as activation functions, the model exhibits the optimal convergence performance. Additionally, raising the values of parameters is able to enhance the convergence performance of the model. Although the convergence rate is influenced by parameter settings and the choice of activation functions, the final converged solution stays consistent.

Fig. 9 presents the convergence of the model under different initial states, with the exception of the initial states being different, all other conditions are identical.

As shown in Fig. 9, the error function can all converge to zero under all initial states, which fully indicates that the convergence of the sznnfrn model is independent of the choice of initial states, further verifying the robust convergence of the model.

7. Applications

7.1. Solve for the value of the equivalent resistance between any two points in fan resistor networks

Traditional approaches to computing the equivalent resistance between arbitrary two points of a resistor network typically require cumbersome analytical derivations. The present study employs the proposed SZNNFRN model to compute the equivalent resistance between arbitrary two points of a fan resistor network.

A comparative analysis was conducted between the equivalent resistance obtained from exact resistance formulas and that computed by the SZNNFRN model, take the selected node (1, 1) in the 3×3 fan resistor network as an example. The parameters γ, λ, and k are set to 10, 3, and 2, respectively, and the SZNNFRN model employed a linear activation function. The comparison results are shown in Fig. 10.

Analysis of Fig. 10 reveals that the equivalent resistance calculated using exact resistance formulas and that solved by the SZNNFRN model exhibit a high degree of consistency, this result fully verifies the accuracy of the SZNNFRN model in calculating the equivalent resistance between arbitrary two points of a fan resistor network. Using the SZNNFRN model to solve for the equivalent resistance between any two nodes in a fan resistor network is a novel method, which not only avoids the complex numerical derivation process required for exact analytical solutions but also maintains high computational accuracy, thereby demonstrating its practicality and applicability in practical applications.

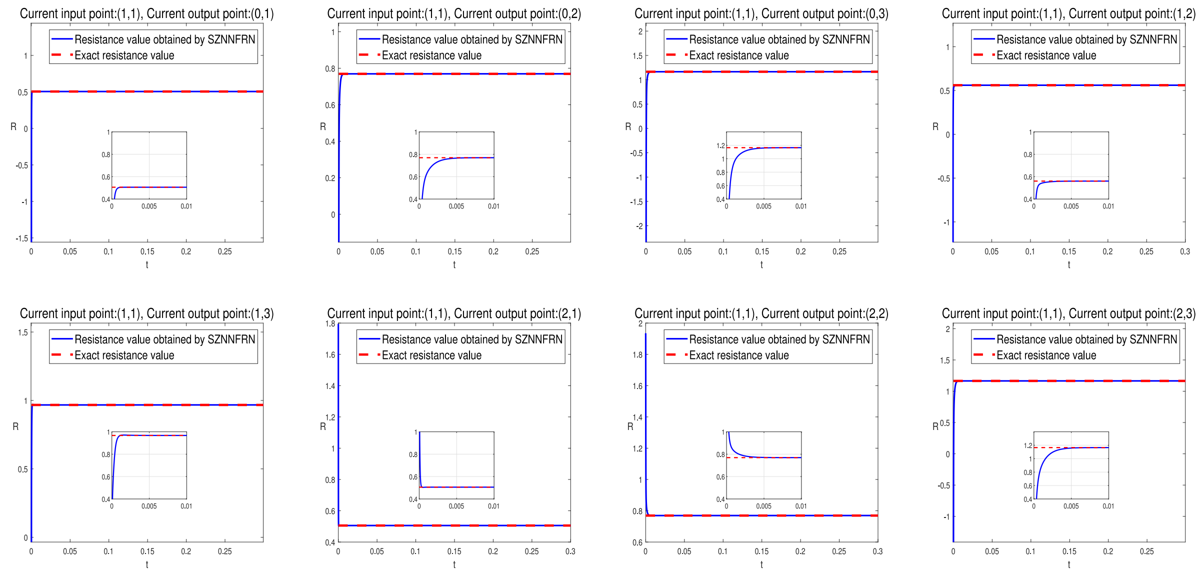


Fig. 10. Comparison curves of equivalent resistance values obtained from exact formula and SZNNFRN model in a 3 × 3 fan resistor network.

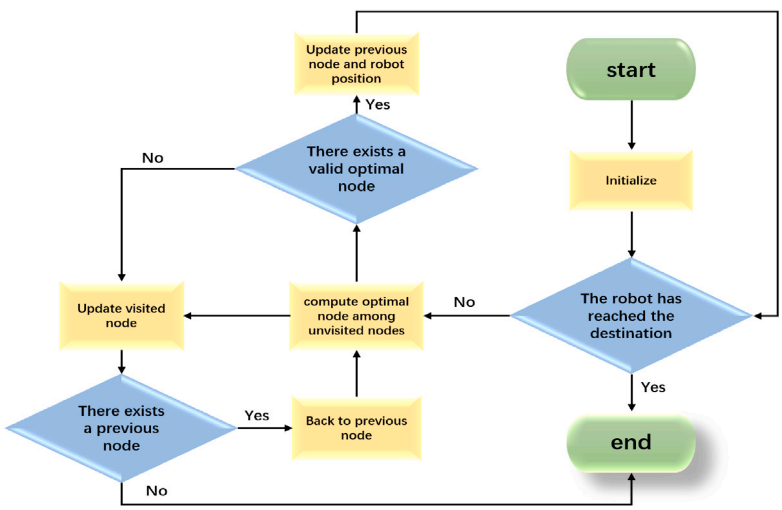


Fig. 11. Flowchart of robot path planning.

7.2. Robot path planning

In the field of artificial intelligence and robotics, robotic path planning occupies an important position [54–57]. Advanced robotic path planning delivers safe, adaptive navigation and higher efficiency. This section implements and evaluates robotic path planning in a fan-shaped curved surface environment. We simulate robot trajectory generation in static and dynamic scenarios using five algorithms — our method, Dijkstra, A*, Floyd and JPS, and present a comprehensive comparison of their computational efficiency.

The core idea of our proposed path-planning algorithm is to exploit the principle of natural potential descent to generate optimal paths. We first apply a grid discretization method to construct a fan-shaped curved surface environment corresponding to the fan resistor network. Using the SZNNFRN model, we obtain the electric potential at each node of the resistor network, and emulate obstacle distributions on the fan-shaped curved surface environment by increasing the potentials of the corresponding nodes. The robot's optimal navigation path is obtained by following the natural descent of the potential. The specific implementation steps of this pathfinding methodology are detailed in the corresponding flowchart Fig. 11 and algorithm 4.

We simulated the path-planning task in a fan-shaped curved surface environment. The fan resistor network was discretized into a 10 × 10 grid, with obstacles precisely placed at the corresponding grid nodes, as shown in Fig. 12.

Algorithm 4 Path planning algorithm

Input: Resistor network scale m, n, robot's start position (S_x, S_y), target position(G_x, G_y), obstacle information (X_i, Y_j), where $i = 1, \cdots, n, j = 1, \cdots, m$.

Step 1: Initialize.

Initialize the map environment, including the start point , end point, visited node set $H\,[x, y]$, current point $(D_x, D_y) = (S_x, S_y)$, previous node $P_0 = (D_x, D_y)$, where k is the number of robot movements and obstacle positions. Initialize the weighted potential field environment under the fan resistor network. Initialize the initial direction from start point to goal point.

if $(S_x, S_y) = (G_x, G_y)$

go to step 4.

else

go to step 2.

Step 2: Compute next node.

Obtain the 8 neighboring points. The optimal node (N_x, N_y) is determined through a weighted evaluation mechanism. This evaluation mechanism comprehensively considers both the node's potential value and distance factors, employing specific weight allocations for each parameter in quantitative calculations to ultimately derive the optimal node.

if Distance $((N_x, N_y),(G_x, G_y))$ < Distance $((D_x, D_y),(G_x, G_y))$

set $P_k = (D_x, D_y)$, (D_x, D_y)=(N_x, N_y).

else

put (N_x, N_y) into visited node. Then recompute the optimal node.

if there is no optimal node exists

put (D_x, D_y) into visited node. backtrack to the previous node P_{k-1}.

else

update the optimal node to the current node and previous node.

Step 3: Check if the destination is reached

if (D_x, D_y)=(G_x, G_y)

then go to step 4.

else

repeat step 2.

Step 4: End.

Output: The path from start position to target position.

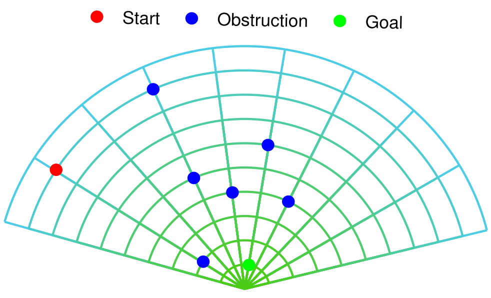


Fig. 12. Initial fan map with obstacles.

The fan resistor network's unique topology ensures that the current inlet and outlet correspond to the nodes of maximum and minimum electric potential, respectively. In the path planning implementation, the starting position corresponds to the maximum-potential node while the destination is assigned to the minimum-potential node. The electric potential on every node in the fan resistor network was determined via SZNNFRN, and the resulting distribution is shown in Fig. 13.

To simulate obstacles in practical environments, we applied weighting to the electric potentials of nodes corresponding to obstacles within the fan-shaped resistor network, thereby generating an adjusted weighted potential distribution as shown in Fig. 14. Based on the weighted potential distribution, the natural descent characteristics of the electric potential are utilized to generate a collision-free path from the input terminal to the output terminal, with the specific path planning results illustrated in Fig. 15.

The optimized navigation path is ultimately mapped onto the fan-shaped curved surface environment incorporating obstacle nodes, with the final configuration presented in Fig. 16.

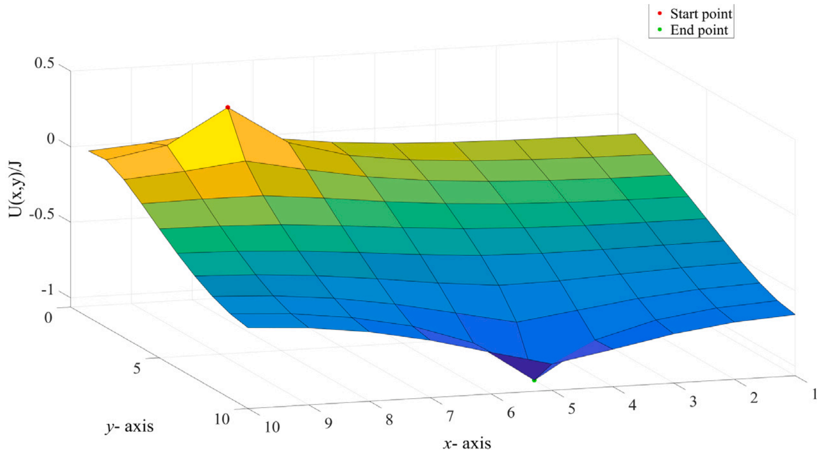


Fig. 13. Potential map of the fan resistor network.

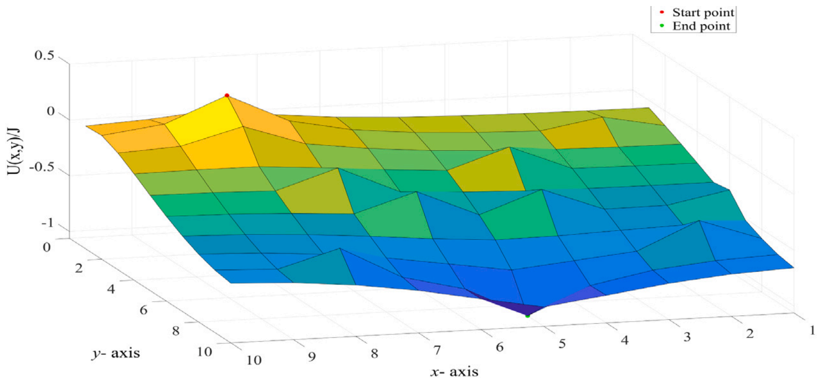


Fig. 14. Weighted potential distribution diagram of fan resistor network.

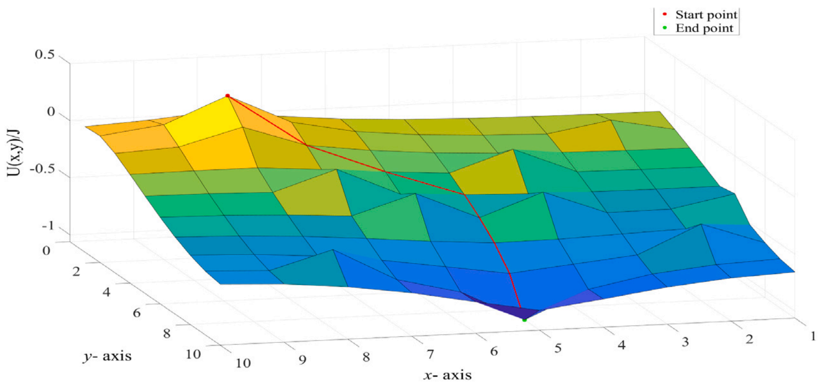


Fig. 15. Path planning within a potential field map with weighted nodes.

For verifying the performance effectiveness of the put forward algorithm, this segment conducts comparative experiments with several classical path planning algorithms. Displaying paths from five algorithms under a common simulated environment (Fig. 17) makes their path characteristics directly comparable.

While all algorithms can complete the path planning task, they exhibit significant differences in efficiency. To compare the efficiency differences among different algorithms, this study conducted tests under completely identical experimental conditions: each algorithm was executed repeatedly 50 times in grids of different sizes, and its average runtime was taken as the final runtime

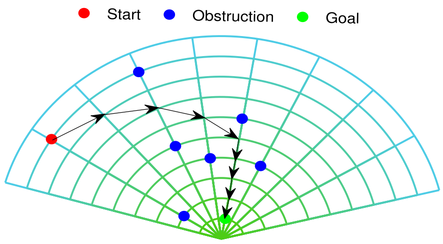


Fig. 16. Robot path planning in fan map with obstacles.

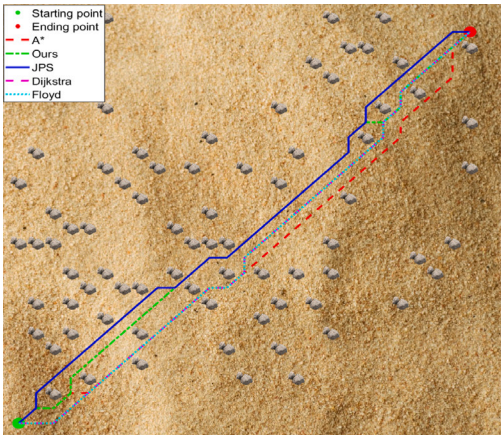


Fig. 17. Path planning diagrams for the five algorithms.

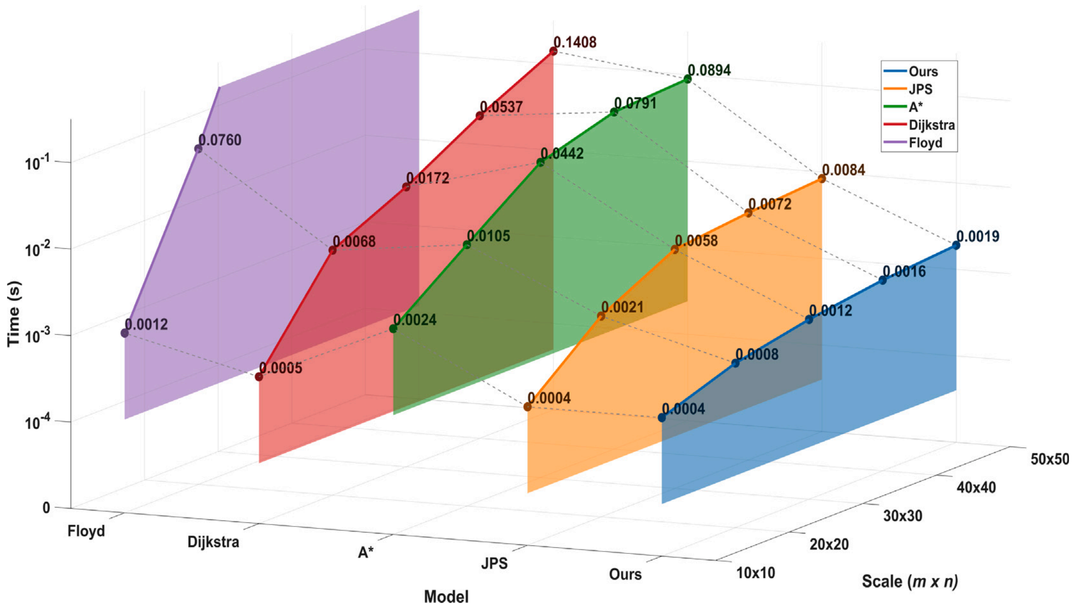


Fig. 18. Efficiency comparison chart of five path planning algorithms.

of the algorithm. The specific experimental results are shown in Fig. 18. If the runtime or memory usage of a certain algorithm exceeds the limit, it is denoted as "NaN", and the corresponding data points are not displayed in the figure.

Analysis results show that in small-scale grids, all algorithms exhibit relatively close efficiency. However, with the increase in grid size, the proposed algorithm demonstrates a significant efficiency advantage, compared with other algorithms, the growth rate of its computation time is significantly slower. When the grid size reaches 50×50, the efficiency improvement of the proposed

Table 2
Comparison of path lengths generated by five path planning algorithms.

	10 × 10	20 × 20	30 × 30	40 × 40	50 × 50
A*	9.4853	23.6274	38.5269	52.0833	66.8112
JPS	9.4853	23.6274	38.5269	52.0833	66.8112
Dijkstra	9.4853	23.6274	38.5269	52.0833	66.8112
Floyd	9.4853	23.6274	38.5269	52.0833	66.8112
Our method	9.4853	23.6274	38.5269	52.6690	67.1041

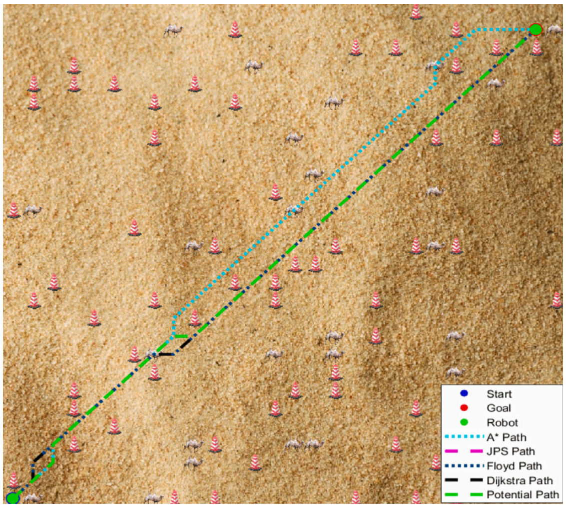


Fig. 19. Dynamic path planning for five algorithms.

algorithm over other algorithms ranges from 5 to 70 times. These results fully confirm that the proposed algorithm possesses excellent scalability, making it particularly suitable for large-scale application scenarios with strict requirements on computation time.

Table 2 compares the path lengths generated by five path planning algorithms under identical experimental conditions. Data analysis results indicate that the proposed method yields the same path length as the A*, JPS, Dijkstra and Floyd algorithms at the network scales of 10 × 10, 20 × 20 and 30 × 30, all achieving the shortest path; a slight deviation in path length from the shortest path is only observed at the 40 × 40 and 50 × 50 network scales. The core design objective of this study is to construct the "fastest path", and the proposed method can guarantee high computational efficiency while attaining the shortest path at most experimental scales. In future research, we will further optimize the path length to realize the synergistic improvement of path length and computational efficiency.

To better approximate real-world application scenarios, this study conducted systematic simulation tests on five path planning algorithms in dynamic environments. Based on observations of obstacle movement characteristics in practical settings, the test environment incorporated both static obstacles and dynamic obstacles — the latter further categorized into three distinct motion patterns: lateral movement, longitudinal movement, and random movement. The simulation results are presented in Fig. 19. (Note: Shown in the figure is the final path. The complete animation of the dynamic path planning process is provided in the supplementary materials).

Although the proposed algorithm exhibits significantly higher efficiency than other algorithms in specific environments, there is still considerable room for improvement in terms of its performance. Future research will focus on two aspects: firstly, extending the application scenarios of this algorithm to more environments, and secondly, further enhancing its performance.

8. Discussion and conjectures

8.1. Avoiding the problem of local potential minimum

In potential field-based path planning methods, the problem of local potential minimum is a common challenge. In the fan resistor network, the potential distribution is relatively smooth, which fundamentally reduces the probability of this local potential minimum problem occurring. Specifically, we can regulate this by adjusting the resistivity and optimize parameters during testing, thereby ensuring that the potential field used for path planning is relatively uniform overall. Furthermore, to address the "pathfinding failure" in potential field-based planning caused by obstacles or dynamic disturbances, our path planning algorithm incorporates direction constraints and a backtracking mechanism. These mechanisms can effectively alleviate the local minimum problem, break through potential local optimal solutions, and ensure that the system always moves toward a globally optimal state.

8.2. The optimality problem in path planning

It is well-known that on a Riemannian manifold, there exists a unique geodesic connecting two nonsingular points. This principle not only ensures the existence of the shortest path but also confirms that, in spacetime, the geodesics representing the trajectories of free fall are exactly the shortest paths (i.e., geodesics) connecting two points on a spatial surface. Based on this, combined with our simulation results, we propose the following conjecture:

Conjecture 1: In a surface resistor network with a fan shape, the potential field formed by the network's own electrical properties is similar to the gravitational field in free fall. The "natural potential descent trajectory" followed by the current from the source point to the sink point is equivalent to the shortest path connecting these two points on the fan-shaped spatial surface.

If the above conjecture is not valid, we can further consider the following physical analogy: Imagine two small balls setting off simultaneously from the same starting point—one travels along a smooth path, while the other moves along an undulating path. In the end, the latter reaches the destination first. This undulating path is precisely the famous "brachistochrone" in physics, which is not the shortest in distance but minimizes the total travel time. By analogy with this, in a fan-shaped surface resistor network, the "natural potential descent trajectory" followed by the current from the source point to the sink point is equivalent to the "brachistochrone" on the fan-shaped spatial surface. This further emphasizes the view that the fastest path is not necessarily the shortest: in variable-speed environments (such as curved surfaces affected by gravity or resistance), paths optimal in terms of time or cost are often achieved through ingenious curve design, rather than the shortest straight-line distance. Based on this, we further conjecture that:

Conjecture 2: Based on the principle of natural potential descent, the route from the start position to the destination position is the "fastest path" on the fan-shaped surface, and enables efficient navigation in complex spatial surface environments.

To summarize, the basis for the optimality of our path planning algorithm lies in the validity of at least one of the two conjectures above. These conjectures are derived from the principle of optimality in natural laws and the results of physical experiments, which provides a new perspective on the optimality problem in path planning. Optimality in path planning is typically defined as the path with the lowest total cost among all feasible paths—and this cost often goes beyond mere Euclidean distance, meaning it is frequently not entirely equivalent to the shortest path. In uniform environments (e.g., open flat planes), the shortest path is indeed optimal, as it minimizes total distance. However, in real-world scenarios where environments are non-uniform (e.g., presence of obstacles, slopes, or speed limits), the shortest path may fall into local traps or high-cost areas, rendering it non-optimal. For instance, robotic navigation's optimal path often trades distance for overall efficiency: autonomous vehicles on rough mountain roads take circuitous, gentler-slope routes to cut fuel consumption, brake wear and travel time; UAV path planning avoids high wind resistance areas, which adds spatial distance but realizes global optimality by lowering energy loss and flight risks.

Our simulation experiments have verified the above conjectures: under various experimental conditions, compared with other algorithms, our proposed path planning algorithm based on natural potential descent has demonstrated higher efficiency and stronger adaptability. Specifically, in large-scale complex scenarios (such as high-dimensional dense obstacle environments or dynamic surface environments), this method can accomplish the path planning task with lower computational complexity while maintaining an asymptotically optimal path, ensuring reliable optimal performance in resource-constrained real-time applications (e.g., autonomous robotic systems).

9. Conclusion

This study proposes the SZNNFRN model for solving the mathematical model of fan dynamic resistor networks. The SZNNFRN model not only extends the set of existing resistor network solution methods but also fills a research gap in the computation of dynamic resistor networks. Based on the distinctive structural features of special matrices, this study develops a class of fast algorithms for the SZNNFRN model. Comparative assessment with current algorithms reveals that the proposed method offers a substantial efficiency advantage, especially in large-scale network computations. The convergence and accuracy of the proposed model are validated through theoretical analysis and numerical experiments. The SZNNFRN model is employed to determine the equivalent resistance between arbitrary node pairs in a fan resistor network. In addition, this study proposes a robot path-planning method for fan-shaped curved-surface environments, which demonstrates a clear efficiency advantage over traditional path-planning algorithms under specific conditions. This research focuses mainly on fan dynamic resistor networks, and thus its applicability to other dynamic resistor networks is still constrained. Future efforts will expand to additional categories of dynamic resistor networks.

CRediT authorship contribution statement

Chen Zhang: Writing – original draft, Validation. **Xiaoyu Jiang:** Conceptualization. **Yanpeng Zheng:** Formal analysis, Data curation. **Zhaolin Jiang:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.physa.2026.131418.

Data availability

The data that has been used is confidential.

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